Chapter 15. The Spectral Theorem

Normal, symmetric and orthogonal operators

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Traditionally the set of eigenvalues of an operator is its *spectrum*, and theorems describing operators by (orthonormal) eigen-decompositions are called spectral theorems. With the tools of Chapters 13–14 we can now go much further than the bare hands of Chapter 9.

15.1 Notation and terminology

Motivation. Transpose and adjoint operators give a single notation T^* for quadratic and Hermitian spaces. Subspaces invariant under both T and T^* make induction on dimension possible, and the conditions $T^* = T$, $T^*T = I$, $TT^* = T^*T$ single out the symmetric, orthogonal and normal operators.

Notation 15.1.1

"$T \in \mathfrak{LM}$" means $T \in \mathfrak{L}(V,V)$ or $T \in \mathfrak{M}_{n,n}(F)$, with $\dim V = n$; a matrix T is regarded as L_T on $V = F^n$. We assume $V \neq 0$.

Notation 15.1.2

(a) For $F = \mathbb{C}$ and a non-degenerate Hermitian space (V,H), T^* means the adjoint $T^\star$; for a matrix A, $T^* = (L_A)^\star$. (b) For a non-degenerate quadratic space (V,B), T^* means the transpose T^t; we usually write T^t to stress the quadratic case.

Remark 15.1.3

(a) Dual maps have done their job of introducing transposes and adjoints and do not appear again. (b) For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ in $(\mathbb{C}^n, H)$, $[(L_A)^\star]_{\mathcal{E}} = \bar{J}^{-1}A^*\bar{J}$ ($J = [H]_{\mathcal{E}}$), which must be distinguished from the adjoint matrix A^*; they agree when $\mathcal{E}$ is orthonormal (e.g. the Hermitian dot product). (c) Likewise $[(L_A)^t]_{\mathcal{E}} = J^{-1}A^tJ$ in (F^n, B), which equals the transpose matrix A^t when $\mathcal{E}$ is orthonormal (e.g. the ordinary inner product).

Observation 15.1.4

If $W \leq V$ is both T-invariant and T^*-invariant, then $(T|_W)^* = T^*|_W$.

Proof. Writing $\langle\,,\rangle$ for H or B: $\langle T|_W w, w'\rangle = \langle Tw, w'\rangle = \langle w, T^*w'\rangle = \langle w, T^*|_W w'\rangle$ for $w, w' \in W$; apply Propositions 14.3.11, 13.9.3 (on W, when $\langle\,,\rangle|_W$ is non-degenerate, e.g. for inner products). ■

Definition 15.1.5 (Special operators)

For $T \in \mathfrak{LM}$: (a) if $T^* = T$, T is **self-adjoint** (**Hermitian**) in the case (V,H), and **symmetric** in the case (V,B); (b) if $T^*T = I$, T is **unitary** (case (V,H)) or **orthogonal** (case (V,B)); (c) if $TT^* = T^*T$, T is **normal**. ($A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is a **normal matrix** if $AA^* = A^*A$.)

Exercise 15.1.6

(a) Self-adjoint (symmetric) operators are normal.
(b) Unitary (orthogonal) operators are normal.

Solution. (a) $TT^* = T^2 = T^*T$.

(b) $T^*T = I$ forces T invertible with $T^{-1} = T^*$, so $TT^* = I = T^*T$. □

15.2 Normal operators

Motivation. On a Hermitian inner product space the operators that are diagonal in some orthonormal basis are exactly the normal ones. As a consequence every unitary matrix is conjugate in $\mathbf{U}(n)$ to a diagonal matrix of numbers of modulus 1.

Notation 15.2.1

In this section $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional space with a Hermitian inner product.

Observation 15.2.2

If V has an orthonormal basis of eigenvectors of T, then T is normal.

Proof. With such $\mathfrak{B}$, $[H]_{\mathfrak{B}} = I$ and $[T]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, so by Proposition 14.3.12 $[T^*]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(\bar\lambda_1, \dots, \bar\lambda_n)$. Diagonal matrices commute. ∎

Observation 15.2.3

T is normal iff $\langle Tv, Tw\rangle = \langle T^*v, T^*w\rangle$ for all v, w.

Proof. If T is normal, $\langle Tv, Tw\rangle = \langle v, T^*Tw\rangle = \langle v, TT^*w\rangle = \langle T^*v, T^*w\rangle$. Conversely, the same chain read backwards gives $\langle v, T^*Tw\rangle = \langle v, TT^*w\rangle$ for all v, w, so $T^*T = TT^*$. ∎

Observation 15.2.4

If T is normal and $\lambda \in \mathbb{C}$, then $T - \lambda I$ is normal.

Proof. $(T - \lambda I)^* = T^* - \bar\lambda I$ (Observation 14.3.15, Exercise 14.3.16), and $(T - \lambda I)(T^* - \bar\lambda I) = TT^* - \bar\lambda T - \lambda T^* + |\lambda|^2 I = (T^* - \bar\lambda I)(T - \lambda I)$. ∎

Observation 15.2.5

If T is normal and $Tv = \lambda v$, then $T^*v = \bar\lambda v$.

Proof. By Observations 15.2.3, 15.2.4: $0 = \langle (T - \lambda I)v, (T - \lambda I)v\rangle = \langle (T^* - \bar\lambda I)v, (T^* - \bar\lambda I)v\rangle$, so $(T^* - \bar\lambda I)v = 0$ by positive definiteness. ∎

Theorem 15.2.6 (Spectral theorem, positive definite Hermitian case)

For $T \in \mathfrak{LM}$ (Notation 15.2.1) the following are equivalent: (1) T is normal; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1) is Observation 15.2.2. Let T be normal; induct on n. Over $\mathbb{C}$, T has a unit eigenvector v, $Tv = \lambda v$. By Gram–Schmidt $V = \langle v\rangle \oplus \langle v\rangle^\perp$; let $W = \langle v\rangle^\perp$.

(i) W is T-invariant: for $u \in W$, $\langle v, Tu\rangle = \langle T^*v, u\rangle = \langle \bar\lambda v, u\rangle = \bar\lambda\langle v, u\rangle = 0$ (Observation 15.2.5).

(iii) W is T^*-invariant: $\langle v, T^*u\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$.

(ii) By Observation 15.1.4, $(T|_W)^* = T^*|_W$, so $T|_W$ is normal. By induction W has an orthonormal basis of eigenvectors of $T|_W$; add v. ∎

So normal operators act, in a suitable orthonormal frame, by scalar multiplication on each coordinate axis.

Definition 15.2.7 (Maximal tori)

With $\mathbf{D}_n(\mathbb{C})$ the invertible diagonal matrices, $\mathbf{T}_{\mathbf{U}(n)} = \mathbf{U}(n) \cap \mathbf{D}_n(\mathbb{C})$ and $\mathbf{T}_{\mathbf{SU}(n)} = \mathbf{SU}(n) \cap \mathbf{D}_n(\mathbb{C}) = \mathbf{T}_{\mathbf{U}(n)} \cap \mathrm{SL}_n(\mathbb{C})$.

Exercise 15.2.8

(a) $\mathbf{T}_{\mathbf{U}(n)} = \{\mathrm{diag}(\lambda_1, \dots, \lambda_n) : |\lambda_i| = 1\}$. (b) $\mathbf{T}_{\mathbf{U}(n)} \approx S^1 \times \cdots \times S^1 \approx \mathbf{U}(1) \times \cdots \times \mathbf{U}(1)$ (n factors). (c) $\mathbf{T}_{\mathbf{SU}(n)} \approx S^1 \times \cdots \times S^1$ ($n-1$ factors).

Solution. (a) A diagonal matrix has orthonormal columns iff $|\lambda_i| = 1$.

(b) $\mathrm{diag}(\lambda_i) \mapsto (\lambda_i)$.

(c) The last entry is determined by $\lambda_n = (\lambda_1 \cdots \lambda_{n-1})^{-1}$ (Exercise 11.9.13). □

Corollary 15.2.9 (Spectral theorem for $\mathbf{U}(n)$)

If $A \in \mathbf{U}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ for some $U \in \mathbf{U}(n)$. Hence $\mathbf{U}(n) = \bigcup_{U \in \mathbf{U}(n)} U\, \mathbf{T}_{\mathbf{U}(n)}\, U^{-1}$.

Proof. In $\mathbb{C}^n$ with the Hermitian dot product, $(L_A)^\star = L_{A^*}$ (Remark 15.1.3(b)), so L_A is unitary, hence normal (Exercise 15.1.6). By Theorem 15.2.6 there is an orthonormal eigenbasis $\mathfrak{B}$. Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} \in \mathbf{U}(n)$ is diagonal (Remark 10.6.4, Proposition 14.1.12), so $[I]^{\mathfrak{B}}_{\mathcal{E}} A [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{T}_{\mathbf{U}(n)}$, and $U = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{U}(n)$ (Exercise 10.6.9(5)). ■

(So every element of $\mathbf{U}(n)$ is conjugate to an element of $\mathbf{T}_{\mathbf{U}(n)}$, and $\mathbf{U}(n)$ is the union of the conjugates of $\mathbf{T}_{\mathbf{U}(n)}$.)

Exercise 15.2.10

If $U \in \mathbf{U}(n)$, then $\alpha U \in \mathbf{SU}(n)$ for some $\alpha \in \mathbb{C}$.

Solution. $|\det U| = 1$, so $\det U = e^{i\theta}$; take $\alpha = e^{-i\theta/n}$, so that $\det(\alpha U) = \alpha^n \det U = 1$ and $\alpha U \in \mathbf{U}(n)$ ($|\alpha| = 1$). □

Corollary 15.2.11 (Spectral theorem for $\mathbf{SU}(n)$)

If $A \in \mathbf{SU}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(n)}$ for some $U \in \mathbf{SU}(n)$; so $\mathbf{SU}(n) = \bigcup_{U \in \mathbf{SU}(n)} U\, \mathbf{T}_{\mathbf{SU}(n)}\, U^{-1}$.

Proof. By Corollary 15.2.9, $U^{-1}AU$ is diagonal unitary with determinant $\det A = 1$, i.e. in $\mathbf{T}_{\mathbf{SU}(n)}$, for some $U \in \mathbf{U}(n)$. Replace U by $\alpha U \in \mathbf{SU}(n)$ (Exercise 15.2.10); $(\alpha U)^{-1} A (\alpha U) = U^{-1}AU$. ■

Exercise 15.2.12

In Corollary 15.2.9, can U always be chosen in $\mathbf{SU}(n)$, even when $A \notin \mathbf{SU}(n)$?

Solution. Yes: replace U by αU as in the proof of Corollary 15.2.11; the conjugate is unchanged. □

Exercise 15.2.13

If $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is normal ($AA^* = A^*A$), then $U^{-1}AU \in \mathbf{D}_n(\mathbb{C}) \cup \{\text{singular diagonal}\}$, i.e. $U^{-1}AU$ is diagonal, for some $U \in \mathbf{SU}(n)$.

Solution. L_A is normal for the dot product, so Theorem 15.2.6 gives an orthonormal eigenbasis; proceed as in Corollary 15.2.9 and Exercise 15.2.12. □

Exercise 15.2.14

For $A = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right) \in \mathbf{SU}(2)$, find $U \in \mathbf{SU}(2)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(2)}$.

Solution. The eigenvalues are $\frac{1\pm i}{\sqrt{2}} = e^{\pm i\pi/4}$, with unit eigenvectors $\frac{1}{\sqrt{2}}(1,1)^t$ and $\frac{1}{\sqrt{2}}(-1,1)^t$. $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ has $\det U = 1$, and $U^{-1}AU = \operatorname{diag}(e^{i\pi/4}, e^{-i\pi/4})$. □

15.3 Symmetric operators

Motivation. Over $\mathbb{R}$ eigenvalues may fail to exist, but symmetric operators always have real eigenvalues and are diagonal in an orthonormal basis. Applied to the matrix of a quadratic form, this diagonalizes quadratic forms by a rotation of axes and characterises positive definiteness by eigenvalues.

Notation 15.3.1

In this section (through Exercise 15.3.11) $F = \mathbb{R}$, $(V, \langle\,,\rangle)$ is an n-dimensional real inner product space, and we write T^t for T^*.

Proposition 15.3.2

(a) A symmetric operator T has a real eigenvalue.
(b) A real symmetric matrix has a real eigenvalue.

Proof. $(V, \langle\,,\rangle)$ is isomorphic to $(\mathbb{R}^n, \text{dot product})$ (§10.5), so we may take $T = L_A$, $A \in \mathfrak{M}_{n,n}(\mathbb{R})$; then $(L_A)^t = L_{A^t}$ and $A = A^t$. View $L_A : \mathbb{C}^n \to \mathbb{C}^n$ with the Hermitian dot product, which extends the dot product of $\mathbb{R}^n$. Take an eigenvector $X \in \mathbb{C}^n$, $AX = \lambda X$, $\lambda \in \mathbb{C}$. Since $A^* = A^t = A$, Observation 10.7.1 gives $\lambda\langle X, X\rangle = \langle AX, X\rangle = \langle X, A^*X\rangle = \langle X, \lambda X\rangle = \bar{\lambda}\langle X, X\rangle$, so $\lambda = \bar{\lambda} \in \mathbb{R}$. ■

So the characteristic polynomial of every symmetric operator splits over $\mathbb{R}$.

Observation 15.3.3

If V has an orthonormal basis of eigenvectors of T, then T is symmetric.

Proof. With such $\mathfrak{B}$, $[B]_{\mathfrak{B}} = I$ and $[T^t]^{\mathfrak{B}}_{\mathfrak{B}} = I^{-1}([T]^{\mathfrak{B}}_{\mathfrak{B}})^t I = [T]^{\mathfrak{B}}_{\mathfrak{B}}$ (Proposition 13.9.4), since diagonal matrices equal their transposes. ■

Theorem 15.3.4 (Spectral theorem, positive definite real case)

For $T \in \mathfrak{LM}$ (Notation 15.3.1) the following are equivalent: (1) T is symmetric; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1) is Observation 15.3.3. For (1)⇒(2), copy the proof of Theorem 15.2.6 with $*$ replaced by ^{t} and no conjugations: Proposition 15.3.2 provides a real unit eigenvector v; $W = \langle v\rangle^\perp$ is T-invariant ($\langle v, Tu\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$); $T|_W$ is symmetric; induct. ■

Corollary 15.3.5 (Spectral theorem for real symmetric matrices)

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ (dot product on $\mathbb{R}^n$) the following are equivalent: (1) A is symmetric; (2) $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A; (3) $U^{-1}AU$ is diagonal for some $U \in \mathbf{O}(n)$.

Proof. Theorem 15.3.4 for L_A, with U the orthogonal matrix whose columns are the orthonormal eigenbasis. ■

Diagonalizing quadratic forms. For $2Q(x_1, \dots, x_n) = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j$ and symmetric $J = (a_{ij})$, $2Q(X) = X^tJX$. Choose $U \in \mathbf{O}(n)$ with $U^{-1}JU = D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Since $U^t = U^{-1}$, with $X = UX'$,

$$2Q(X) = X^tUDU^{-1}X = (U^{-1}X)^tD(U^{-1}X) = \lambda_1(x_1')^2 + \cdots + \lambda_n(x_n')^2 :$$

the quadratic form is **diagonalized**. (The point is $U^t = U^{-1}$.)

Example 15.3.6

For $J = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $2Q(x, y) = 2x^2 + 2xy + 2y^2$, and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) J \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right)$. Substituting $x = \frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$, $y = -\frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$ gives $2x^2 + 2xy + 2y^2 = (x')^2 + 3(y')^2$.

Example 15.3.7

The curve $2x^2 + 2xy + 2y^2 = 4$: in the orthonormal eigenbasis $\mathfrak{B} = \{v, w\} = \{\frac{1}{\sqrt{2}}(1, -1)^t, \frac{1}{\sqrt{2}}(1, 1)^t\}$ with $U = [I]^{\mathcal{E}}_{\mathfrak{B}}$, $X = UX'$ means $X = x'v + y'w$. The equation becomes $(x')^2 + 3(y')^2 = 4$: an ellipse with axes along v and w, i.e. a standard ellipse rotated clockwise by 45°.

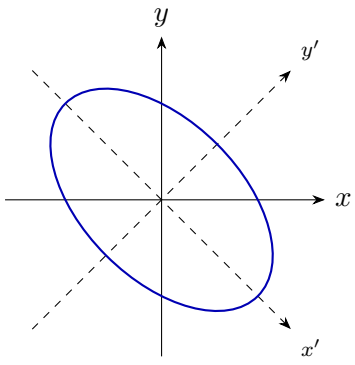


Figure. $2x^2 + 2xy + 2y^2 = 4$ *is the ellipse* $(x')^2 + 3(y')^2 = 4$ *in the rotated frame.*

Exercise 15.3.8

$x^2 + xy + y^2 - x - y + \frac{1}{3} \geq 0$ for all real x, y.

Solution. Substituting $x = u + \frac{1}{3}$, $y = v + \frac{1}{3}$ kills the linear terms and the constant: the expression becomes u^2+uv+v^2. Its matrix $\left(\begin{smallmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{smallmatrix}\right)$ has eigenvalues $\frac{3}{2}, \frac{1}{2} > 0$, so $u^2+uv+v^2 = \frac{3}{2}(u')^2+\frac{1}{2}(v')^2 \geq 0$. □

Exercise 15.3.9

Sketch (a) $2x^2 - 2\sqrt{2}xy + 3y^2 = 4$;
(b) $x^2 - 6xy + y^2 = 2$;
(c) $11x^2 + 24xy + y^2 = 15$.

Solution. (a) $J = \begin{pmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 3 \end{pmatrix}$ has eigenvalues $1, 4$, with eigenvectors $(\sqrt{2}, 1)$, $(-1, \sqrt{2})$: the ellipse $(x')^2 + 4(y')^2 = 4$. (b) $J = \begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix}$ has eigenvalues -2 (eigenvector $(1, 1)$) and 4 (eigenvector $(-1, 1)$): the hyperbola $4(y')^2 - 2(x')^2 = 2$. (c) $J = \left(\begin{smallmatrix} 11 & 12 \\ 12 & 1 \end{smallmatrix}\right)$ has eigenvalues 19 (eigenvector $(3, 2)$) and -7 (eigenvector $(-2, 3)$): the hyperbola $19(x')^2 - 7(y')^2 = 15$. □

Exercise 15.3.10

Why are the high-school conics (with centre at the origin) exactly ellipses (including circles) and hyperbolas, plus parabolas?

Solution. After rotating axes (Corollary 15.3.5) a quadratic equation $X^tJX + \ell(X) + c = 0$ has quadratic part $\lambda_1(x')^2 + \lambda_2(y')^2$. Completing squares in the variables with $\lambda_i \neq 0$: if $\lambda_1\lambda_2 > 0$ one gets an ellipse (or a point or nothing), if $\lambda_1\lambda_2 < 0$ a hyperbola (or a pair of lines), and if exactly one $\lambda_i = 0$ a parabola (or degenerate lines). □

Exercise 15.3.11

For a finite-dimensional real quadratic space (V, B) with basis $\mathfrak{B}$, the following are equivalent: (1) B is positive definite; (2) all eigenvalues of $J = [B]_{\mathfrak{B}}$ are positive.

Solution. $B(v, v) = X^tJX$ with $X = [v]_{\mathfrak{B}}$. Diagonalize $J = UDU^t$ with $U \in \mathbf{O}(n)$ (Corollary 15.3.5): $X^tJX = \sum \lambda_i (x_i')^2$ with $X' = U^tX$ ranging over all of $\mathbb{R}^n$. This is positive for all $X' \neq 0$ iff all $\lambda_i > 0$. (This answers Question 13.1.14.) □

Exercise 15.3.12

For a finite-dimensional Hermitian space (V, H) with basis $\mathfrak{B}$, the following are equivalent: (1) H is positive definite; (2) all eigenvalues of $J = [H]_{\mathfrak{B}}$ are positive reals.

Solution. $H(v, v) = X^tJ\bar{X} = \overline{\bar{X}^t\bar{J}X}$ with $X = [v]_{\mathfrak{B}}$. $\bar{J}$ is self-adjoint, hence unitarily diagonalizable with real eigenvalues (normal, Theorem 15.2.6, and Proposition 15.3.2's argument), which are those of J (conjugates of real numbers). Then $H(v, v) = \sum \lambda_i |x_i'|^2$, positive for all $X \neq 0$ iff all $\lambda_i > 0$. □

15.4 Orthogonal operators

Motivation. A real orthogonal operator need not be diagonalizable (rotations), but it splits V into mutually orthogonal invariant lines and planes. Hence every element of $\mathbf{SO}(n)$ is conjugate in $\mathbf{SO}(n)$ to a block-diagonal matrix of plane rotations: $\mathbf{SO}(n)$ is a mixture of copies of $\mathbf{SO}(2)$.

Here $F = \mathbb{R}$.

Proposition 15.4.1

Every $T \in \mathfrak{LM}$ over $\mathbb{R}$ has an invariant subspace W with $\dim W = 1$ or 2.

Proof. Factor the minimal polynomial $m_T = p_1 \cdots p_s$ into monic irreducibles in $\mathbb{R}[t]$. Some $0 \neq v$ has $p_1(T)v = 0$: if $s = 1$ take any $v \neq 0$; if $s \geq 2$, minimality gives w with $v = p_2(T) \cdots p_s(T)w \neq 0$. Irreducible real polynomials have degree 1 or 2. If $p_1 = t + c$, then $Tv = -cv$; if $p_1 = t^2 + at + b$, then $T^2v = -aTv - bv$. In either case $W = \langle v, Tv \rangle$ is T-invariant, of dimension 1 or 2. ■

Theorem 15.4.2 (Spectral theorem for real orthogonal operators)

If V is a finite-dimensional real inner product space and T is orthogonal, there are subspaces $W_1, \ldots, W_s$ of V with (1) $V = W_1 \oplus \cdots \oplus W_s$; (2) each W_i is T-invariant; (3) $W_i \perp W_j$ for $i \neq j$; (4) $\dim W_i \in \{1, 2\}$.

Proof. Induction on $\dim V$. Take W as in Proposition 15.4.1; $V = W \oplus W^\perp$ (Gram–Schmidt). $W^\perp$ is T-invariant: T is injective, so $TW = W$, and every $w' \in W$ is Tw for some $w \in W$; for $v \in W^\perp$, $\langle w', Tv \rangle = \langle Tw, Tv \rangle = \langle w, T^tTv \rangle = \langle w, v \rangle = 0$. $T|_{W^\perp}$ is orthogonal on the inner product space $W^\perp$; apply induction. ■

Corollary 15.4.3 (Spectral theorem for $\mathbf{O}(n)$)

For $A \in \mathbf{O}(n)$ there are $A_i \in \mathbf{SO}(2)$ and $U \in \mathbf{O}(n)$ with
$A \in \mathbf{SO}(n)$, $n = 2m$: $U^{-1}AU = \operatorname{diag}(A_1, \ldots, A_m)$;
$A \in \mathbf{SO}(n)$, $n = 2m + 1$: $U^{-1}AU = \operatorname{diag}(1, A_1, \ldots, A_m)$;
$A \notin \mathbf{SO}(n)$, $n = 2m$: $U^{-1}AU = \operatorname{diag}(-1, 1, A_1, \ldots, A_{m-1})$;
$A \notin \mathbf{SO}(n)$, $n = 2m + 1$: $U^{-1}AU = \operatorname{diag}(-1, A_1, \ldots, A_m)$.

Proof. (i) Apply Theorem 15.4.2 to L_A on $\mathbb{R}^n$ with the dot product; collecting orthonormal bases of the W_i gives an orthonormal basis $\mathfrak{B}$, $[I]_{\mathcal{E}}^{\mathfrak{B}} \in \mathbf{O}(n)$ (Observation 9.3.6(5)), and $[I]_{\mathcal{E}}^{\mathfrak{B}} A [I]_{\mathfrak{B}}^{\mathcal{E}} = [L_A]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ with B_i of size 1 or 2.

(ii) $\operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ iff each $B_i \in \mathbf{O}(1)$ or $\mathbf{O}(2)$. So (a) $B_i = \pm 1$ if $\dim W_i = 1$;

(b) if $B_i \in \mathbf{O}(2) - \mathbf{SO}(2)$, B_i is a reflection, so $U_i^{-1}B_iU_i = \operatorname{diag}(-1, 1)$ for some $U_i \in \mathbf{O}(2)$ (§9.4); inserting U_i we may assume $B_i = \operatorname{diag}(-1, 1)$.

(iii) Reordering $\mathfrak{B}$ (a permutation matrix, which lies in $\mathbf{O}(n)$) gives $\operatorname{diag}(\pm 1, \ldots, \pm 1, C_1, \ldots, C_r)$ with $C_j \in \mathbf{SO}(2)$. Now $I_2 = R_0$ and $-I_2 = R_\pi$ lie in $\mathbf{SO}(2)$, so pairs of 1's or of -1's can be absorbed into rotations. At most one 1 and at most one -1 remain, and since $\det A = (-1)^{\#(-1)}$, reordering once more yields the four stated forms according to the parity of n and whether $\det A = \pm 1$. ■

Definition 15.4.4 (Maximal torus of $\mathbf{SO}(n)$)

(a) For $n = 2m$: $\mathbf{T}_{\mathbf{SO}(n)} = \{\operatorname{diag}(A_1, \ldots, A_m) : A_i \in \mathbf{SO}(2)\}$. (b) For $n = 2m + 1$: $\mathbf{T}_{\mathbf{SO}(n)} = \{\operatorname{diag}(1, A_1, \ldots, A_m) : A_i \in \mathbf{SO}(2)\}$.

Exercise 15.4.5

(a) $\mathbf{T}_{\mathbf{SO}(n)} \leq \mathbf{SO}(n)$.
(b) $\mathbf{T}_{\mathbf{SO}(n)} \approx \mathbf{SO}(2) \times \cdots \times \mathbf{SO}(2) \approx S^1 \times \cdots \times S^1$ ($m = \lfloor n/2 \rfloor$ factors).

Solution. Block-diagonal products and inverses are computed blockwise, and $\det = \prod \det A_i = 1$; $\operatorname{diag}(\ldots, A_i, \ldots) \mapsto (A_i)$ is an isomorphism, and $\mathbf{SO}(2) \approx S^1$ (Example 11.9.19). □

Corollary 15.4.6 (Spectral theorem for $\mathbf{SO}(n)$)

If $A \in \mathbf{SO}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SO}(n)}$ for some $U \in \mathbf{SO}(n)$. Hence $\mathbf{SO}(n) = \bigcup_{U\in\mathbf{SO}(n)} U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$.

Proof. By Corollary 15.4.3, $U^{-1}AU = \operatorname{diag}(1, R_{\theta_1}, \dots, R_{\theta_m}) \in \mathbf{T}_{\mathbf{SO}(n)}$ for some $U \in \mathbf{O}(n)$ (for even n ignore the leading 1; assume $n \geq 2$). If $U \in \mathbf{SO}(n)$ we are done. Otherwise let $S = \operatorname{diag}(1, I_2, \dots, I_2, \operatorname{diag}(-1, 1)) \in \mathbf{O}(n) - \mathbf{SO}(n)$, which flips one coordinate of the last plane. By Observation 9.5.9, $\operatorname{diag}(-1,1)R_\theta \operatorname{diag}(-1,1) = R_{-\theta}$, so $(US)^{-1}A(US) = \operatorname{diag}(1, R_{\theta_1}, \dots, R_{\theta_{m-1}}, R_{-\theta_m}) \in \mathbf{T}_{\mathbf{SO}(n)}$, and $US \in \mathbf{SO}(n)$. ■

Exercise 15.4.7

In Corollary 15.4.3, can U be chosen in $\mathbf{SO}(n)$ even when $A \notin \mathbf{SO}(n)$?

Solution. Yes. In the normal forms for $A \notin \mathbf{SO}(n)$ there is a block -1; replacing the corresponding basis vector u by $-u$ multiplies $\det U$ by -1 and leaves $U^{-1}AU$ unchanged (u spans a 1-dimensional invariant subspace). □

So orthogonal operators of Euclidean space, in a suitable orthonormal frame, rotate planes and flip at most one axis: "$\mathbf{SO}(n)$ is a mixture of $\mathbf{SO}(2)$'s", and the elements $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ and $\operatorname{diag}(R_\theta, R_\eta) \in \mathbf{SO}(4)$ of Examples 9.6.7 and 9.6.10 are in fact the typical ones.

Exercise 15.4.8

For $A \in \mathbf{SO}(n)$ below, find $U \in \mathbf{SO}(n)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SO}(n)}$: (a) $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right)$; (b) $A = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{smallmatrix}\right)$.

Solution. (a) $\det A = 1$. The eigenvalue 1 has unit eigenvector $u_1 = \frac{1}{\sqrt{2}}(1,1,0)^t$, and $A = -I$ on $u_1^\perp = \langle u_2, u_3\rangle$ with $u_2 = \frac{1}{\sqrt{2}}(-1,1,0)^t$, $u_3 = e_3$. With $U = (u_1, u_2, u_3)$ ($\det U = 1$), $U^{-1}AU = \operatorname{diag}(1, R_\pi)$. (b) Eigenvalue 1: $u_1 = \frac{1}{\sqrt{2}}(1,1,0,0)^t$, $u_2 = \frac{1}{\sqrt{2}}(0,0,1,-1)^t$; eigenvalue -1: $u_3 = \frac{1}{\sqrt{2}}(1,-1,0,0)^t$, $u_4 = \frac{1}{\sqrt{2}}(0,0,1,1)^t$. With $U = (u_1, u_2, u_3, u_4)$, $U^{-1}AU = \operatorname{diag}(R_0, R_\pi)$; $\det U = 1$ (if it were -1, replace u_4 by $-u_4$, which keeps $R_\pi = -I_2$). □

15.5 Epilogue

Motivation. Without positive definiteness the spectral theorems fail: a vector may be orthogonal to itself, and symmetric or self-adjoint operators need not be diagonalizable. Over $\mathbb{C}$, Schur's triangularization survives. And the perp theorem, proved without Gram–Schmidt, applies verbatim to binary codes.

In §§15.2–15.4 Gram–Schmidt was essential. In a non-degenerate quadratic space, if $Tv = \lambda v$ with $B(v,v) = 0$, the decomposition $\langle v\rangle \oplus \langle v\rangle^\perp$ makes no sense.

A false proof. "Let (V, H) be a non-degenerate finite-dimensional Hermitian space and T normal. Take an eigenvector v, put $\bar{V} = V/\langle v\rangle$, let $\bar{T}$ be induced, define $\bar{H}(\bar{u}, \bar{w}) = H(u, w)$, and induct." The flaw: $\bar{H}$ is not well defined unless $v \perp V$, which non-degeneracy forbids. The statement itself is false:

Example 15.5.1

For $J = \operatorname{diag}(1,-1)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ in $(\mathbb{R}^2, B^J_{\mathcal{E}})$, L_A is symmetric (Exercise 13.9.6(b)), but $\phi_A(t) = (t-2)^2$ and $A \neq 2I$, so L_A is not diagonalizable.

Example 15.5.2

For $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ in $(\mathbb{C}^2, H^J_{\mathcal{E}})$: $H^J_{\mathcal{E}}(A(x,y)^t, (z,w)^t) = (x+y)\bar{w} + y\bar{z} = H^J_{\mathcal{E}}((x,y)^t, A(z,w)^t)$, so L_A is self-adjoint, yet not diagonalizable.

(Orthogonal and unitary geometry continue beyond the second-year course.)

Theorem 15.5.3 (Schur)

If $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space, then for every $T \in \mathfrak{LM}$ there is an orthonormal basis $\mathfrak{B}$ with $[T]^{\mathfrak{B}}_{\mathfrak{B}}$ upper triangular.

Proof. Take a unit eigenvector v of T^*, $T^*v = \mu v$. $V = \langle v\rangle^\perp \oplus \langle v\rangle$, and $\langle v\rangle^\perp$ is T-invariant: for $w \perp v$, $\langle v, Tw\rangle = \langle T^*v, w\rangle = \mu\langle v, w\rangle = 0$. By induction $\langle v\rangle^\perp$ has an orthonormal basis $\mathfrak{C}$ with $[T|_{\langle v\rangle^\perp}]^{\mathfrak{C}}_{\mathfrak{C}}$ upper triangular; $\mathfrak{B} = \mathfrak{C} \cup \{v\}$, with v placed last, works. (It suffices to treat $T = L_A$, where Observation 10.7.1 provides T^*.) ∎

Over $\mathbb{C}$ only, this proof is simpler than those of §7.3 and §12.5, but those work over any algebraically closed field.

Field-independent proofs and coding theory. $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$ with $\bar{1}\cdot\bar{1} = \bar{1}$ (and all other products $\bar{0}$) is a field.

Definition 15.5.5 (Binary codes)

A $\mathbb{Z}_2$-subspace C of $(\mathbb{Z}_2)^n$ is a **linear binary code of length** n; with the ordinary inner product $B^I_{\mathcal{E}}$ on $(\mathbb{Z}_2)^n$, $C^\perp$ is the **dual code**.

Theorem 15.5.6

If C is a linear code in $(\mathbb{Z}_2)^n$, then $\dim C + \dim C^\perp = n$; so if $\dim C = m$, $|C| = 2^m$ and $|C^\perp| = 2^{n-m}$.

Proof. $B^I_{\mathcal{E}}$ is non-degenerate on $(\mathbb{Z}_2)^n$ ($\det I = 1 \neq 0$), so this is the perp theorem, Theorem 13.5.11 or 13.8.11, whose proofs work over any field. Note that Gram–Schmidt is unavailable: e.g. $B^I_{\mathcal{E}}((\bar{1},\bar{1})^t, (\bar{1},\bar{1})^t) = \bar{1} + \bar{1} = \bar{0}$. ∎